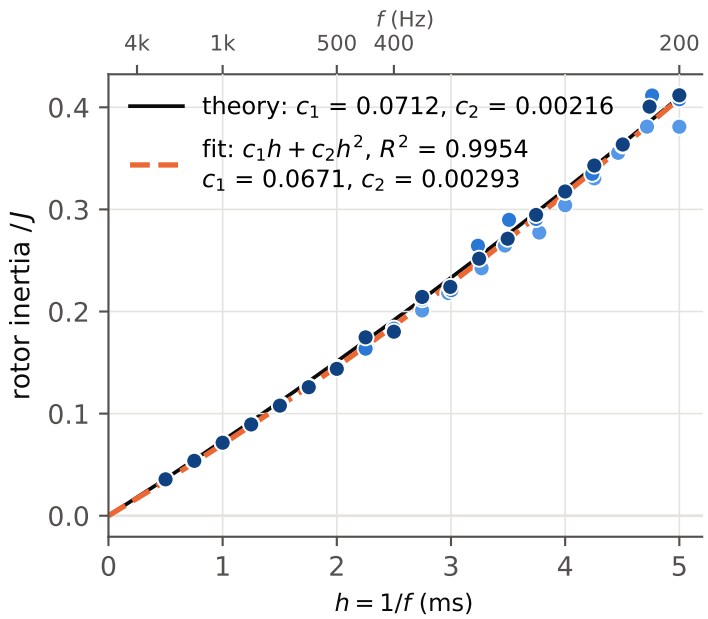
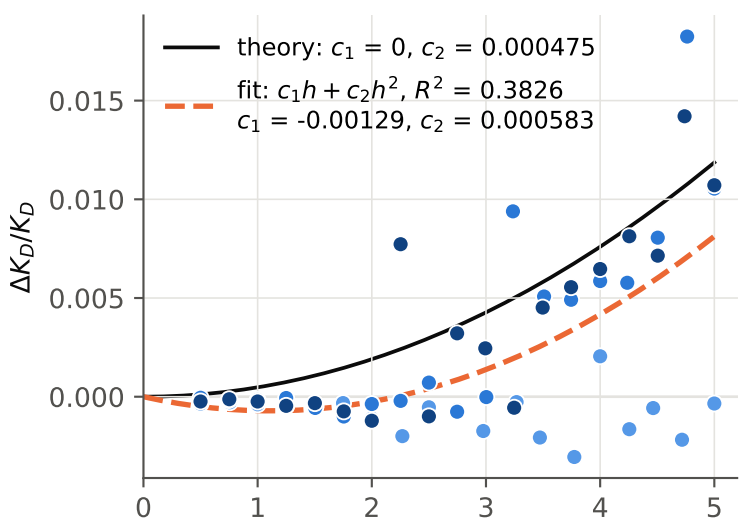
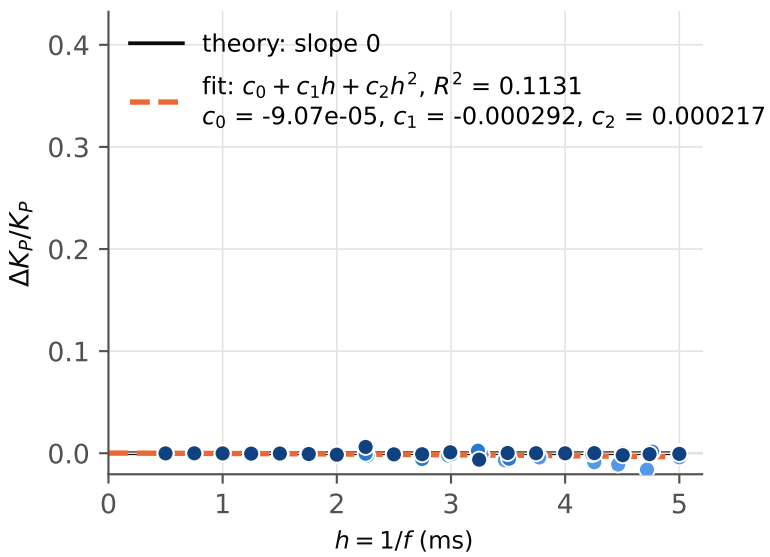
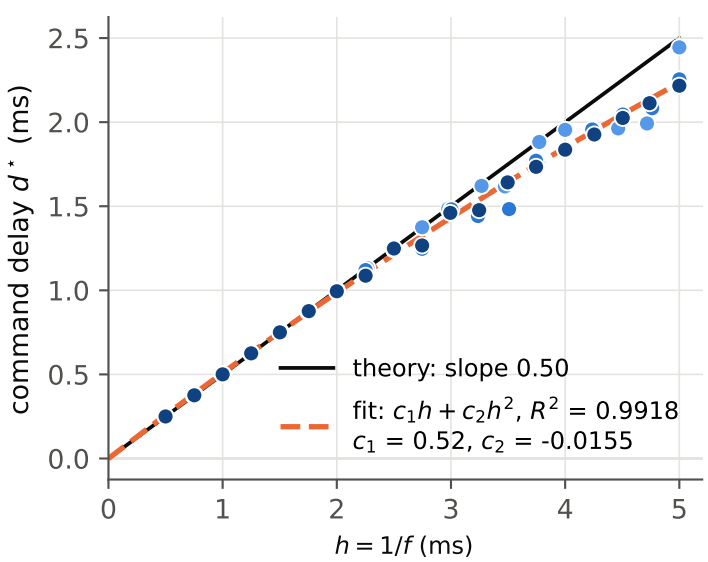
$f_{ctrl} \approx$ 50 Hz  $f_{ctrl} \approx$ 100 Hz  $f_{ctrl} \approx$ 200 Hz

## explicit PD + TGS
fitted: rotor inertia, $K_P$, $K_D$, $d$

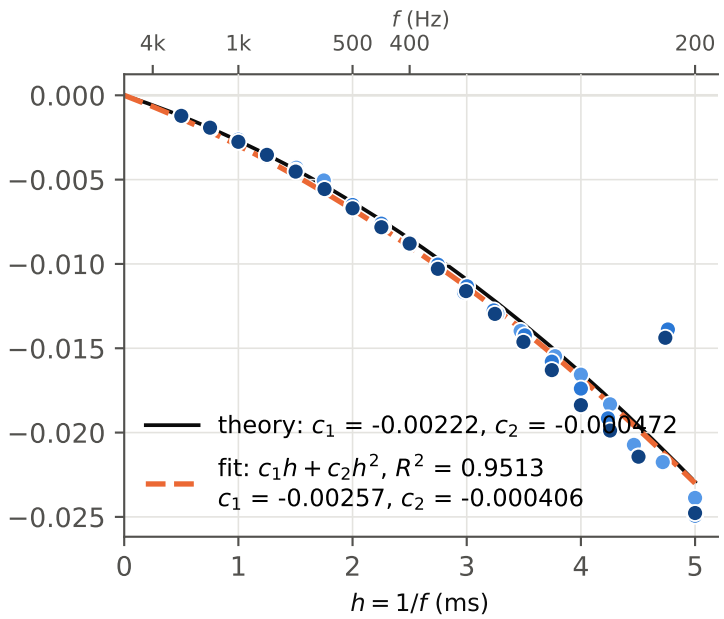


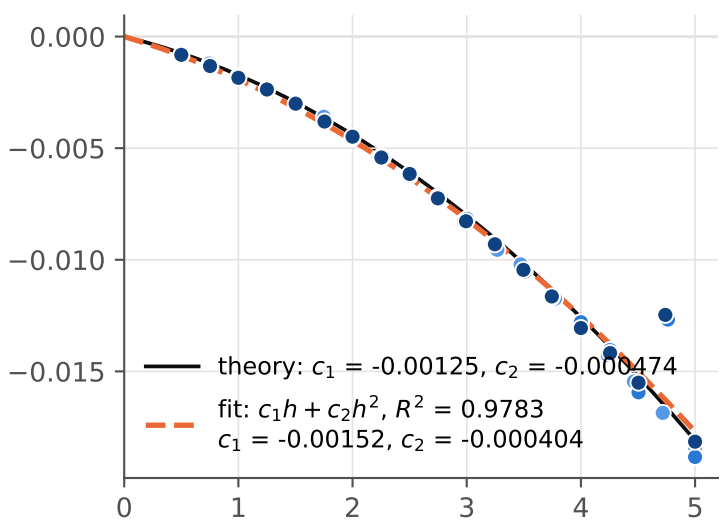



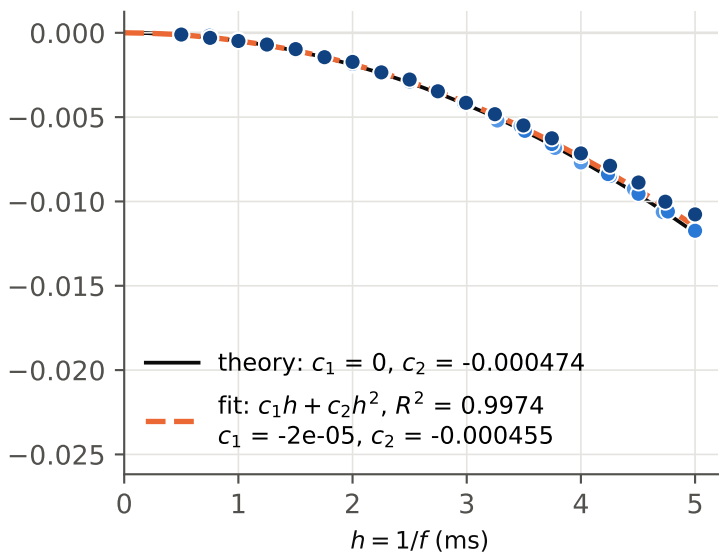



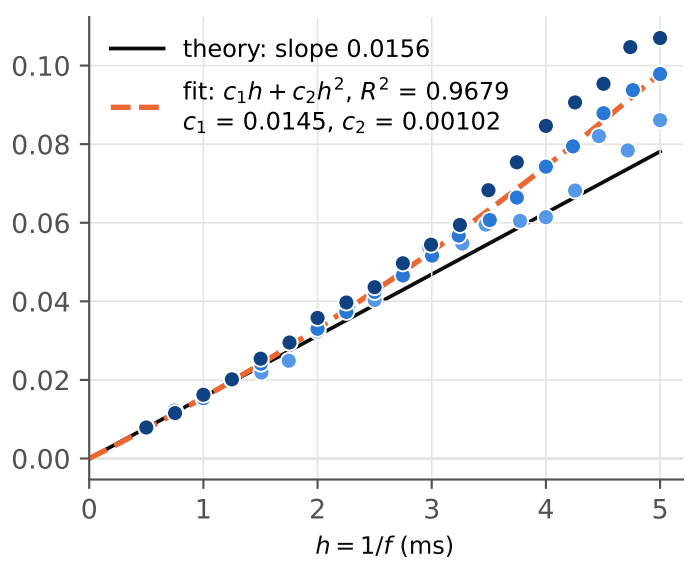




## PhysX drive + TGS
fitted: rotor inertia, $K_P$, $K_D$, $d$

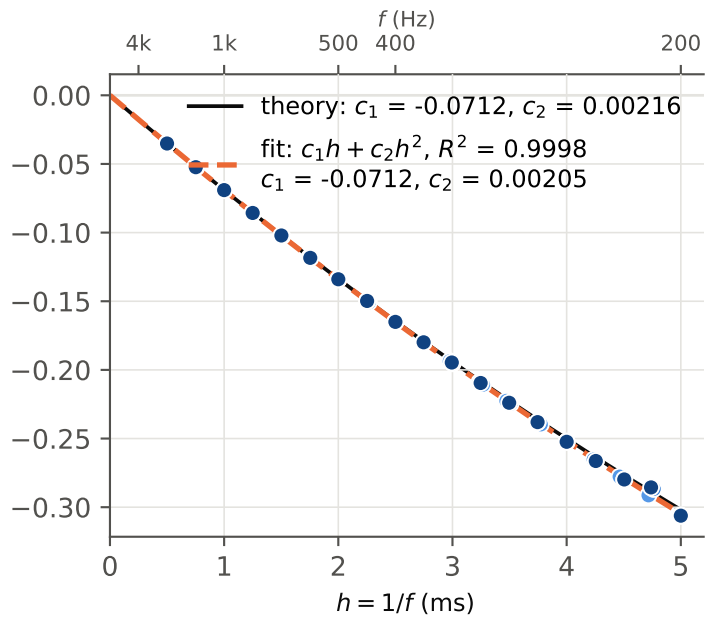


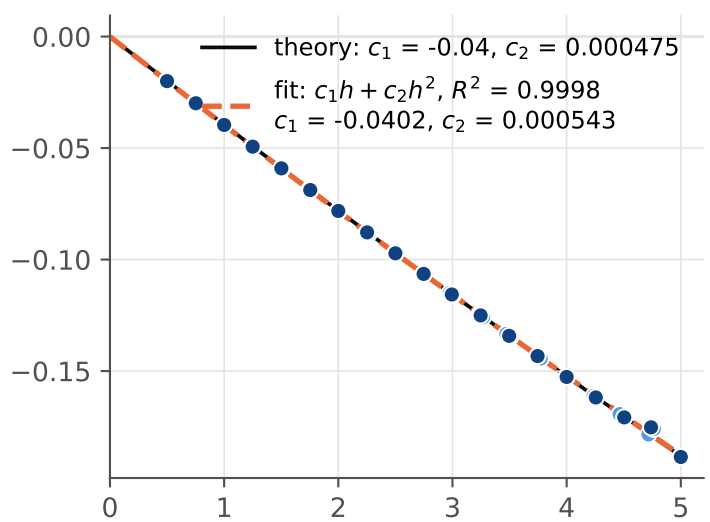



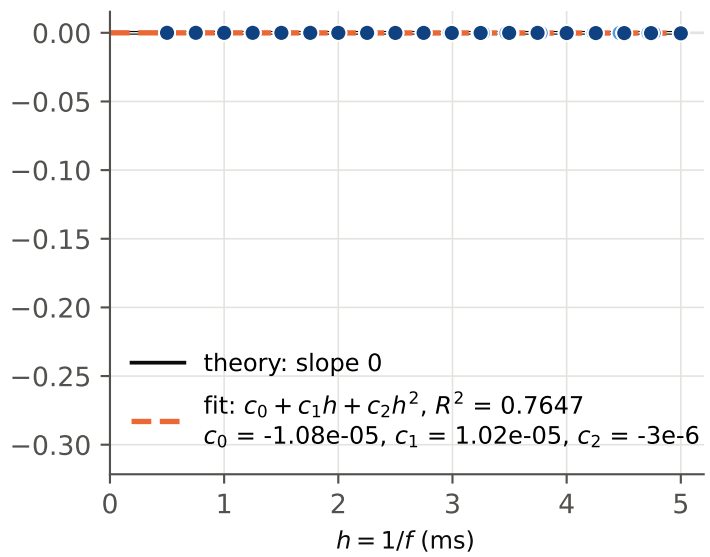



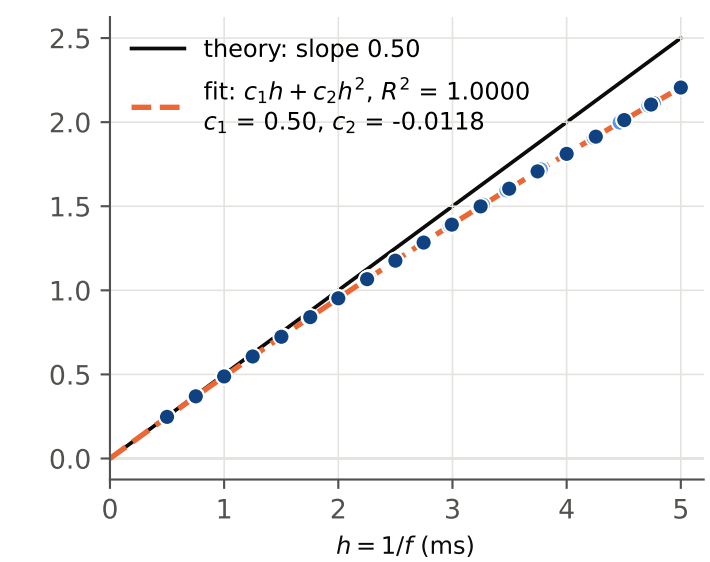




## PhysX drive + PGS
fitted: rotor inertia, $K_P$, $K_D$, $d$




Isaac Sim 5.1 / PhysX, single pendulum, on the same $h$ grid as the MuJoCo figure.
First-order theory (solid); fit (dashed), $c_1h + c_2h^2$ through the origin, so $c_1$ is the measured first-order slope.
An explicitly held torque makes PhysX behave as semi-implicit Euler; its own drive is an implicit spring whose damping is $K_D + \delta K_P$, run at $\delta = h/32$ under TGS and $\delta = h$ under PGS.